PAGE No.
DATE / /

19|6|26

Python Programming
Assignment - 3

1) What is a user defined function? Write a function to accept two numbers and return ~~thei~~ their ~~mu~~ multiplication.

Ans: A user defined function is a function created by user to perform a specific task and which is a reusable ~~code~~ code block of structure.

Multiplication.py

```
print ("Enter first number : ", No1)
No1 = int(input())

print( "Enter second number : ", No2)
No2 = int(input())

def multiply (a, b):
        Multiplication = a * b
        return Multiplication

Ans = multiply (No1, No2)

print ("Multiplication is : ", Ans)
```

DATE

Ans :

2) What is the difference between :

- Function with parameters

→ The function which takes some value in inputs inside the parenthesis (). It uses that information to change the output

ex :-

```
def Hello (name):
    print ("Hello, " + name + "!")

Hello ("Marvellous")
```

- Function without parameters

→ The function which does not takes any inputs. Its parentheses () are empty.

example :-

```
def Hello ():
    print ("Hello, Friend!")

Hello ()
```

Ans

3) Predict the output:

```
def fun ():
    x = 10
    print (x)

fun ()
print (x)
```

Explain the reason:

PAGE No.
DATE / /

Ans : As we know Python runs the code line-by-line.
So, in the above code we have defined fun() function and in that function x have a ~~value i~~ integer value 10 and in the next line it prints its value.
This process will ~~haven~~ happen the fun() function is executed and after executing it ~~destroys it be me~~ there is no memory allocation and also fun() is a local variable. it means it only exists in inside function.

And print(x) is outside the function at the last line so it does not have any idea what it is because fun() is not a Global variable. That's why it will give NameError: name 'x' is not defined.

4) Write a function that does not return anything but prints a message. Explain the default return value of such a function.

Ans:

```
def greetings():
    print ("Jay Ganesh...")

greetings()
```

The default return value of such function will be 'None'. Because, if the user doesn't assign any return value the python by ~~dee~~ default gives ~~a~~ it a return value i.e. None. we can find the default return value by below code.

DATE

```
def fun():
    print("Jay Ganesh ...")

result = fun()
print(" Return value is : ", result)
```

5) What is the difference between:
print()
return
Explain with function example:

Ans: The difference between print() & return is one of the most common points of confusion.
The core difference is

- print() : It simply displays text on your screen so you can see it. It does not save or pass that data to the rest of your program. Once it shows on the screen & it is gone.

- return : It is for the computer. It hands a value back to your program so you can store it in a variable, use it in math equations, or pass it to other functions. It doesn't show anything on the screen by default.

for example -
- print() :

```
def add(a,b):
    result = a+b
    print(result)
```

PAGE No. DATE / /

```
add (5,5)

Ans = add (5,5)
```

- return :

```
def add (a,b)
    result = a + b
    return result

add (5, 5)

Ans = add (5,5)
print (Ans)
```

6) Write a program to display :

- Data type

Ans:
```
no = 11
print ("Type of no is : ", type(no))
```

- Memory address

Ans:
```
no = 11
print ("Memory address of no is : ", id(no))
```

- Size in bytes of a variable entered by the user.

Ans:
```
import sys

print (" Enter a variable : ")
user_input = input()
```

```
try:
    value = int(user_input)
    datatype = "int"
except ValueError:
    try:
        value = float(user_input)
        data_type = "float"
    except ValueError:
        value = user_input
        data_type = "str"

size = sys.getsizeof(value)

print("Variable entered is: {value}")
print("Data type is: {data_type}")
print("Size is :: {size} bytes")
```

7) Predict the output:

```
a=5
print(type(a))      # <class 'int'>
a = 5.5
print(type(a))      # <class 'float'>
a= "Python"
print(type(a))      # <class 'str'>
```

What Python features does this demostrate?

→ This demonstrate that Python is Dynamically Typed.

8) Explain why the following code works without declaration:

x = 100

compare this with C or Java.

Ans: This code works without declaration. because Python is Dynamically Typed Language. In Python, we do not have to declare data type of a variable because it automatically determines it's datatype based on the value given to it at runtime. ~~bes~~ In above example, interpreter instantly recognises 100 as an integer and sets up x accordingly.

Comparing with C or Java, they both are Statically Typed Language. we have to declare the data type so that compiler can know the data type of the variable. Below example based on above one

ex:-

```
int x = 100;    // we have specify 'int' first
```

9) What is the difference between:

x = 10

x = "Ten"

Is this allowed? Why?

Ans: The difference is

x=10 : The variable x points to an integer object (int) with the value of 10.

x = "Ten" : The variable x points to an string object (str) with the text value "ten".

This is allowed because Python is dynamically typed language.

PAGE No.
DATE / /

~~In Python,~~ languages like C++ or Java; variables are like fixed boxes that can only hold one type of data (like only integers).
In Python, variables are just name tags that points to objects in memory.
a) When we write x=10, Python creates an integer object 10 in memory and sticks the name tag x on it.
b) When we write x="Ten", Python simply detaches the name tag x from the number 10 and sticks it onto a completely new string object "Ten".

10) Explain how Python manages memory automatically. Why does the programmer not need to explicitly allocate or free memory?

Ans: When we create a variable, ~~it~~ it automatically allocates a spot in Heap section. When we stop using that variable, it automatically throws it away and clears the memory space.
In other programming language like C & C++ user have to request memory space ~~and it~~ by giving specific ~~no~~ number. Like if ~~you~~ we want to enter any 20 character string, ~~the~~ CPU will give the memory which will be store that perticular 20 characters. no ~~exta~~ extra memory will be provided. But comparing in Python there is not limit of ~~mo~~ memory size giving also there is an method which is Garbage collector which wipes out all the memory which is not been used by the program.